Twitter Topic Classification & Pattern Mining

Original tweets — Dataset · 3
351

After cleaning — Dataset · 3
300

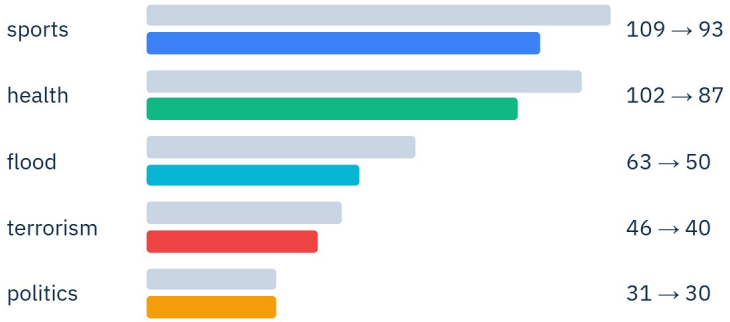
Topic classes — Classes · 3
5

Best test accuracy — Metrics · 5
86.67%

Class distribution

Classes · 3

Grey = original 351 · colour = cleaned 300



Signature words per topic

Dataset · 3

Highest-weight words learned by the SVM

flood flood, kerala, home, damage
health health, care, existing, mental
politics house, politics, white, deputy
sports sports, nadal, kaepernick, usopen
terrorism terrorism, anniversary, america

Input: tweet text → Output: topic label

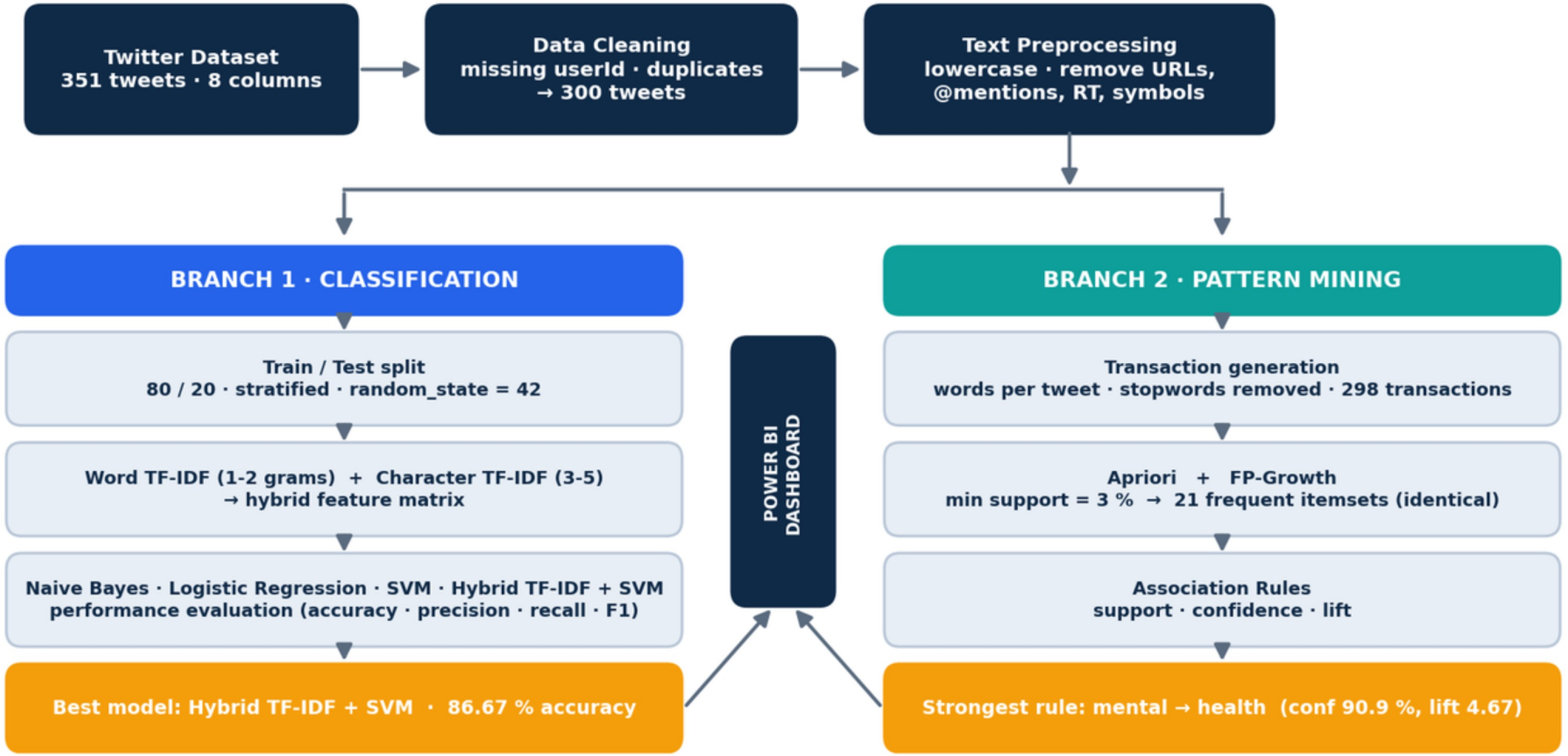
Input / Output · 3

Input tweet (test set)	Predicted	Actual
Flood ins rise faced by many incl Sandy victims – Hard @FEMA homework ahead	**flood**	flood
Hunt for West Wing leakers...	**politics**	politics
Metro Council approve MLS stadium project at the Fairgrounds	**sports**	sports
Today, we remember the horrific tragedy of the 9/11 Attacks in New York and Pennsylvania	**politics**	terrorism

Test set: 60 tweets · 52 correct · 8 wrong

Architecture diagram · 3

System Architecture: Twitter Topic Classification & Pattern Mining



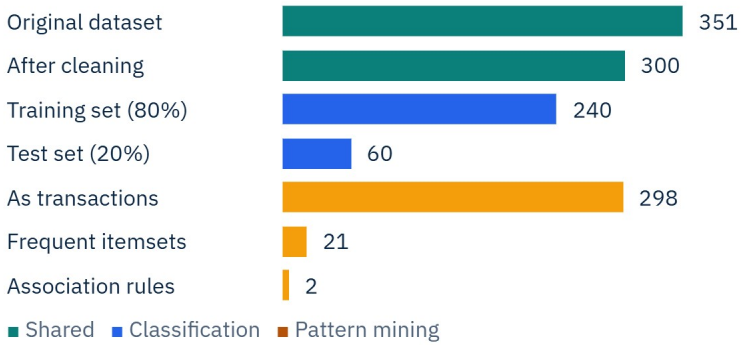
Twitter Topic Classification & Pattern Mining

Architecture diagram | Data pipeline

Branch filters the steps and the stage chart. Topic filters the split chart and the tweet table.

Items at each pipeline stage

Architecture · 3



Branch: Shared | Classification | Pattern mining

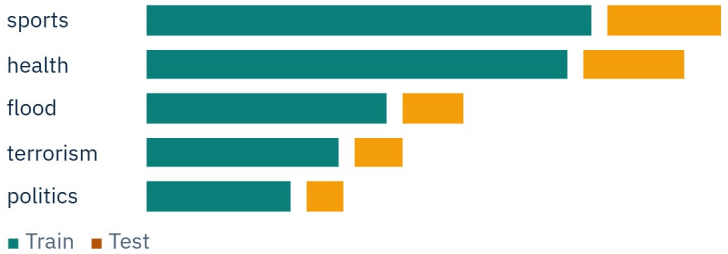
Topic: sports | health | flood | terrorism | politics

Input / Output · 3

#	Step	What happens	Tool
1	**Collect tweets**	351 tweets across 5 topics	CSV file
2	**Clean data**	Remove missing and duplicate tweets, leaving 300	pandas
3	**Preprocess text**	Lowercase; remove URLs, @mentions, RT, symbols	Python regex
4	**Split data**	80/20 stratified: 240 train, 60 test	scikit-learn
5	**Extract features**	Word TF-IDF plus character TF-IDF	TfidfVectorizer
6	**Train models**	NB, Logistic Regression, SVM, Hybrid SVM	scikit-learn

Train / test split by topic

Chart · 3



Pre-processing: raw tweet → cleaned text

Pre-processing · 3

Raw tweet	Cleaned text	Topic
@realDonaldTrump America believes Bob Woodward above your flood of lies.	america believes bob woodward above your flood of lies	**flood**
@Belief_Kay @A_sane_ Live on SABC SPORTS https://t.co/nm1HUVA6as	live on sabc sports	**sports**
Flood ins rise faced by many incl Sandy victims – Hard @FEMA homework ahead #floodinsurance	flood ins rise faced by many incl sandy victims hard homework ahead floodinsurance	**flood**

Twitter Topic Classification & Pattern Mining

Minimum support (slicer)

2% | 3% | 5%

Frequent itemsets (Apriori = FP-Growth) — Pattern · 3

21

Association rules (conf ≥ 50%) — Pattern · 3

2

Strongest lift — Pattern · 3

4.67

Association rules at 3% support

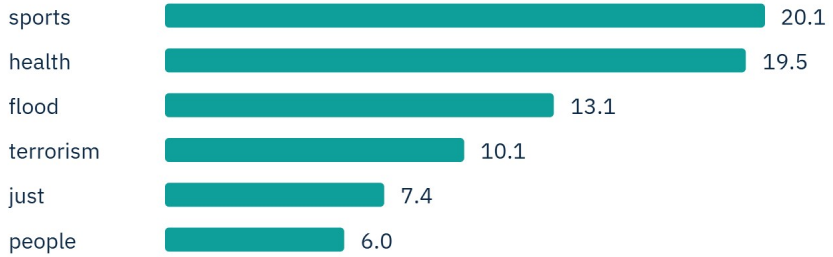
Pattern matching · 3

Rule	Support	Confidence	Lift
mental → health	3.36%	90.91%	4.67
care → health	4.03%	80.00%	4.11

Lift above 1 means the words appear together more often than chance.

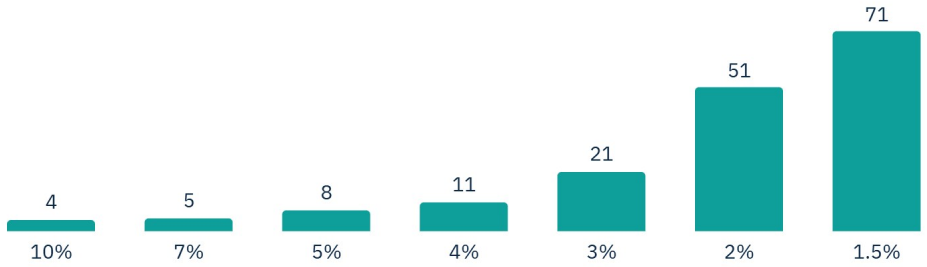
Most frequent single items (support %)

Pattern matching · 3



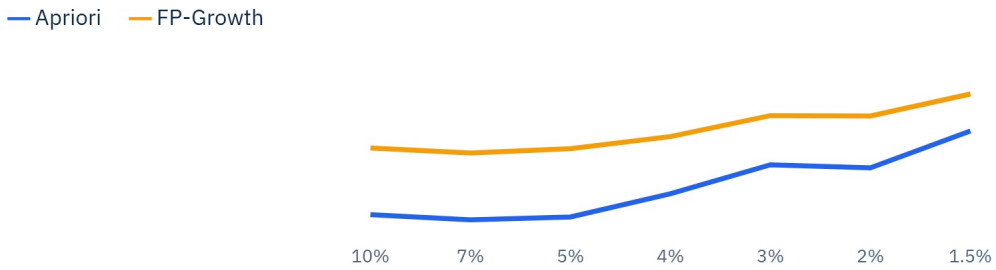
Itemsets found vs minimum support

Chart · 3



Apriori vs FP-Growth runtime (ms)

Pattern · 3



Both algorithms return identical itemsets at every level. At 300 tweets each finishes in a few milliseconds.

Test set (60 tweets): accuracy and F1 (%)

Comparison · 5



Naive Bayes	68.3 · 60.9
Logistic Regression	80.0 · 75.3
SVM	85.0 · 84.7
Hybrid TF-IDF + SVM	**86.7 · 86.1**

Dark bar = accuracy · light bar = weighted F1

Cross-validation accuracy (5-fold × 5 repeats, 300 tweets)

Validation test · 5

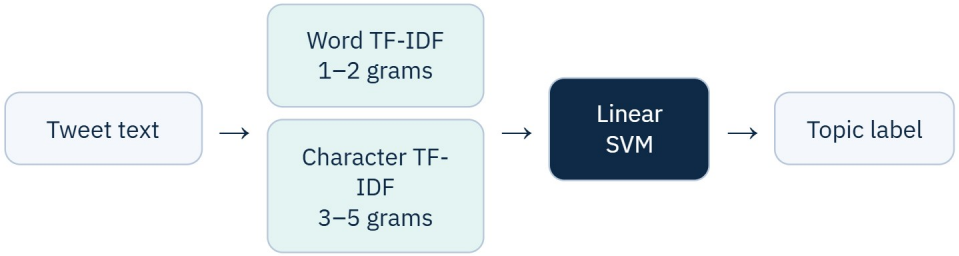


Naive Bayes	66.7 ± 3.0
Logistic Regression	75.8 ± 2.6
SVM	81.3 ± 3.4
Hybrid TF-IDF + SVM	**84.1 ± 3.3**

Black line = ± 1 standard deviation across 25 runs. Every tweet is tested, none leaks into training.

Hybrid TF-IDF + Linear SVM

Classification algorithm · 5



Words capture meaning; character patterns catch spelling variants and hashtags.

Innovation: adding character TF-IDF to the SVM

Innovation · 5

Test set (60 tweets)

85.00 → 86.67

+1.67 points

Cross-validation (300 tweets)

81.3 → 84.1

+2.8 points

The hybrid beat the plain SVM in 19 of 25 validation runs (4 lost, 2 tied). Wilcoxon p = 0.0003.

Accuracy — Metrics · 5

86.67%

Precision (weighted) — Metrics · 5

87.27%

Recall (weighted) — Metrics · 5

86.67%

F1 score (weighted) — Metrics · 5

86.06%

Confusion matrix: Hybrid TF-IDF + SVM (60 test tweets)

Chart · 3

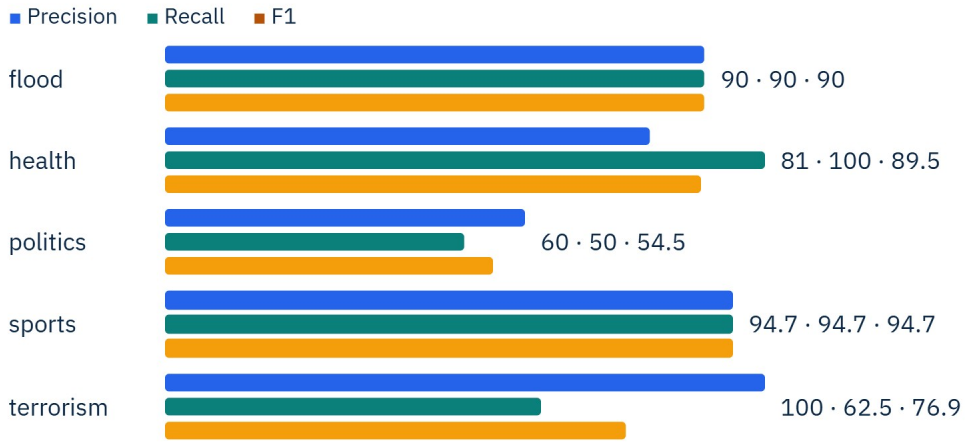
Naive Bayes | Logistic Reg. | SVM | **Hybrid TF-IDF + SVM**

	flood	health	politics	sports	terror.
flood	9	0	0	1	0
health	0	17	0	0	0
politics	1	2	3	0	0
sports	0	0	1	18	0
terrorism	0	2	1	0	5

Rows = actual topic · columns = predicted topic · 52 of 60 on the diagonal

Per-class precision, recall and F1 (%)

Metrics · 5



Politics is the weakest class: only 6 test tweets, mostly confused with health.